FinWiseAI

AI-Powered Personal Finance Assistant

Project Documentation

Submitted by: Astha Singh
Email: astha.singh231006@gmail.com
Class: Second Year B.Tech — Artificial Intelligence & Machine Learning
College: Dr. D. Y. Patil College of Engineering and Innovation, Pune, Maharashtra
Academic Year: 2026–2027

1. Abstract

FinWiseAI is a web-based personal finance application that helps users, especially students, record their income and expenses, understand where their money goes, and receive simple, practical guidance. After creating an account and signing in, the user adds income and expenses on a dashboard that shows total income, total expenses, available balance and a financial health score out of 100, along with spending grouped by category. A Financial Summary page shows the savings rate and a category analysis, a Financial Recommendations page lists practical suggestions based on the recorded data, a Loan Calculator estimates the monthly EMI, total interest and total payment, and a Credit Score page explains credit standing for educational purposes. The AI Advisor module, powered by Google's Gemini model, answers free-text questions about budgeting, spending and saving using the user's recorded finances.

In the screen recording of the running application, every module loaded and behaved as described, the dashboard, summary and recommendations stayed consistent with the recorded entries, and the Loan Calculator's result matched an independent calculation. The credit score shown is illustrative, and all guidance is educational and is not professional financial advice.

2. Introduction

Managing money well starts with knowing how much comes in, how much goes out and which categories take the largest share. Many students do not track this, and ready-made finance tools are often too complex or are spread across separate apps. FinWiseAI brings tracking, analysis, recommendations, a loan calculator, credit education and an AI advisor into one clean interface with a single navigation bar. It is an educational project: its figures are based on the data the user enters, and its advice is general guidance, not a decision by a bank or a certified financial adviser.

3. Problem Statement

Many young people have no simple way to see their income, expenses and savings together, and they rarely get personalised guidance when they need it. Spreadsheets take effort to maintain, and generic finance advice does not reflect a person's own numbers. The proposed application records the user's income and expenses, turns them into clear summaries and a health score, and uses an AI advisor to answer questions with reference to those recorded figures.

4. Objectives

- Provide secure account creation and sign-in so that each user works with their own finance dashboard.
- Let users record income (amount and source) and expenses (amount, category and description).

- Show total income, total expenses, available balance and a financial health score on one dashboard.
- Break spending down by category and present a Financial Summary with the savings rate.
- Generate short, practical recommendations from the recorded data.
- Offer a Gemini-powered AI Advisor that answers budgeting and saving questions using the user's data.
- Provide a Loan Calculator that gives the monthly EMI, total interest and total payment.
- Provide a Credit Score page that explains credit standing and the factors behind it.
- Present all of this in a clean, easy-to-use interface with consistent navigation.

5. Scope

Core features: account registration and login, dashboard with income and expense entry, spending by category, Quick AI insights, Financial Summary, Financial Recommendations (Tips), AI Advisor, Loan Calculator (under Calculators) and the Credit Score page.

Outside the current scope: automatic bank integration (data is entered manually), real credit-bureau score retrieval (the Credit page shows an illustrative score), real lending decisions, and professional financial advice. Further ideas are listed under Future Scope.

6. Technology Stack

Layer	Technology and use
Frontend	HTML5 and CSS3 for the pages; Jinja2 templates to show user data; a responsive card layout with a gradient indigo-to-teal colour theme and a top navigation bar.
Backend	Python with the Flask web framework: routes for registration, login, dashboard, summary, tips, advisor, calculators and credit pages; flash messages such as "Account created. Please log in."
Database	SQLAlchemy ORM with a relational database (SQLite for development; PostgreSQL-ready) to store users, income and expense records.
AI	Google Gemini API, used by the AI Advisor to answer questions using the user's recorded financial snapshot.
Deployment	Runs locally on the Flask development server; Ngrok can be used to give the local server a public URL for demonstration.
Tools	Visual Studio Code, Git, a web browser for testing, and screen recording for the demonstration.

7. Requirements

- A laptop or desktop computer with a modern web browser (the layout also adapts to smaller screens).
- Python 3 with Flask, SQLAlchemy and the Google Gemini client library installed.
- A valid Gemini API key, kept outside the source code (for example in an environment variable).
- An internet connection for the AI Advisor; the other modules work with the local server and database.
- Ngrok (optional) to share the running application through a public link.

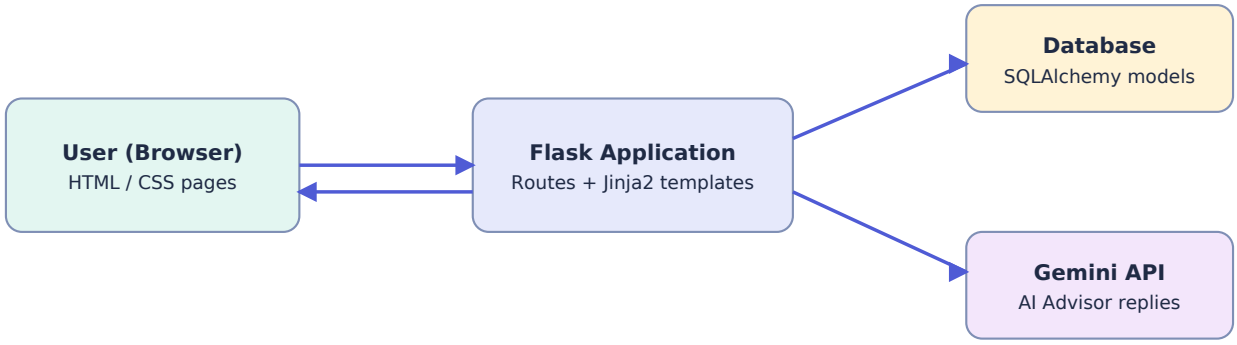
8. System Modules

Module	What it does	Main inputs / outputs
Register and Login	Creates an account and signs the user in; shows a confirmation banner after registration.	Username, email, password (6+ characters). Output: sign-in page and dashboard access.
Dashboard	“Your money at a glance”: summary cards, income and expense forms, spending by category, Quick AI insights and an Ask AI Advisor button.	Income: amount, source. Expense: amount, category, description. Output: Total Income, Total Expenses, Available Balance, Financial Health (out of 100).
Financial Summary	A clear snapshot of the recorded finances with a category analysis.	Output: Income, Expenses, Balance, Savings Rate and amount per category.
Financial Recommendations (Tips)	Practical suggestions based on the recorded activity, with a shortcut to ask Gemini.	Output: a short list of tips, for example the largest expense category.
AI Advisor	Gemini-powered chat-style advice on budgeting, spending and saving, using the user’s recorded data.	Input: a question. Output: a written response with practical steps and an educational disclaimer.
Loan Calculator	Estimates monthly payments and total borrowing cost.	Input: loan amount, annual interest rate (%), years. Output: Monthly EMI, Total Interest, Total Payment.
Credit Score	Credit education page with an illustrative score and the common factors behind a score.	Output: score with standing label and a short explanation.

9. Workflow

1. The user opens the application and creates an account with a username, email and password. 2. After the confirmation banner, the user signs in and lands on the dashboard. 3. The user adds income and expenses; the dashboard cards, balance and category breakdown update from the stored records. 4. The user opens Summary and Tips to review the savings rate, category analysis and recommendations. 5. The user asks a question in the AI Advisor; the application sends the question together with the recorded financial snapshot to Gemini and displays the reply. 6. The user can test borrowing costs in the Loan Calculator and read about credit standing on the Credit page. 7. The user logs out when finished.

10. System Architecture



The browser sends requests to Flask; Flask reads and writes income and expense records through SQLAlchemy, builds each page with Jinja2 and, for the AI Advisor, sends the question and the user’s financial snapshot to the Gemini API and returns the answer to the page.

11. Data Design

The application stores each user's records in a database. The fields below are the ones collected by the forms in the application.

Record	Fields	Notes
User	Username, email, password	Password must be at least 6 characters; records are tied to the signed-in user.
Income	Amount (₹), source	Source is free text, for example "Salary".
Expense	Amount (₹), category, description	Category is chosen from a drop-down (Food, Transport and Bills were used in the demonstration); description is optional.

Calculations shown on the pages

- **Available Balance** = Total Income − Total Expenses (for example ₹0 − ₹2,090 shows ₹-2090.00; ₹0 − ₹12,000 shows ₹-12000.00).
- **Savings Rate** is shown on the Financial Summary page (0.0% in the demonstration, where no income was recorded).
- **Financial Health** is a score out of 100 calculated from the recorded data. It showed 60/100 for a new account with no records and 40/100 once expenses were recorded with no income.
- **Spending by category** adds up the expenses of each category and shows them as labelled bars.
- **EMI** = $P \times r \times (1 + r)^n \div ((1 + r)^n - 1)$, where P is the loan amount, r is the annual rate ÷ 12 ÷ 100 and n is the tenure in years × 12. Total Payment = EMI × n and Total Interest = Total Payment − P.

12. Implementation

Each page is a Jinja2 template that shares one navigation bar (Dashboard, Summary, AI Advisor, Tips, Calculators, Credit, Logout) and one visual theme: white cards on a soft blue background, indigo and teal accents, small uppercase section labels and large page headings. The dashboard summary cards use coloured top borders (green for income, pink for expenses, blue for balance and purple for health) so that each figure is easy to tell apart.

Registration and login use separate forms; after a successful registration Flask flashes a confirmation message on the sign-in page. The dashboard forms post income and expense entries to the server, which saves them and refreshes the cards and the category bars. The Summary and Tips pages are generated from the same stored records, so their numbers always agree with the dashboard.

The AI Advisor page shows a "Gemini connected" status, a text box and a Get Financial Advice button. The replies in the demonstration refer to the user's own totals, balance, savings rate and top categories, which shows that the recorded snapshot is passed to the model with the question. Every reply begins with a reminder that the assistant gives general educational information, is not a certified financial adviser and will not ask for sensitive credentials. The Loan Calculator takes three values and displays the monthly EMI, total interest and total payment in three result cards.

13. Testing

The application was exercised in a single screen-recorded session (2 October 2026). The table lists what was done and what the application displayed.

Test case	Expected result	Actual result
Open the registration page	Create-account form is shown	“Create account” form with Username, Email and Password (6+ characters) fields (Figure 1)
Create an account	Account is saved and the user is sent to sign-in	Sign-in page with the banner “Account created. Please log in.” (Figure 2)
Sign in and open the dashboard	Dashboard opens with the navigation bar	“Your money at a glance” dashboard loaded (Figure 3)
New account with no records	All money cards show zero	Income ₹0.00, Expenses ₹0.00, Balance ₹0.00, Financial Health 60/100 (Figure 3)
Add expenses: Food ₹5,000, Transport ₹4,000, Bills ₹3,000	Totals, balance, health score and category bars update	Total Expenses ₹12000.00, Balance ₹-12000.00, Health 40/100; category bars shown (Figures 5, 6)
Open Financial Summary	Numbers agree with the dashboard	Income ₹0.00, Expenses ₹12000.00, Balance ₹-12000.00, Savings Rate 0.0%; Bills ₹3000.00, Transport ₹4000.00, Food ₹5000.00 (Figure 7)
Open Financial Recommendations	Tips refer to the recorded data	Five tips, including “Your largest recorded expense category is Food at ₹5,000.00.” (Figures 8, 9)
AI Advisor with expenses ₹2,090 (all Food) and no income	Personalised, practical reply	Reply quoted expenses ₹2,090, income ₹0 and balance ₹-2,090 and gave three steps (Figures 10, 11)
AI Advisor with expenses ₹12,000 and no income	Reply reflects the recorded categories	Reply quoted ₹12,000 expenses, Food ₹5,000, Transport ₹4,000, Bills ₹3,000 and gave three steps (Figure 12)
Loan Calculator: ₹1,00,000 at 7.5% for 6 years	EMI about ₹1,729 (independent calculation)	Monthly EMI ₹1729.01; Total Interest ₹24488.81; Total Payment ₹124488.81 (Figures 13, 14)
Open the Credit page	Score with standing and explanation	Score 742 with “Good standing” and a note on common scoring factors (Figure 15)

Not demonstrated in the recording: submitting an income entry, wrong-password and empty-field handling, other loan values, logout, and layout on a phone-sized screen. These should be tested before final submission.

14. Results and Screenshots

The figures below are taken from the running application, in the order a user would meet the screens.

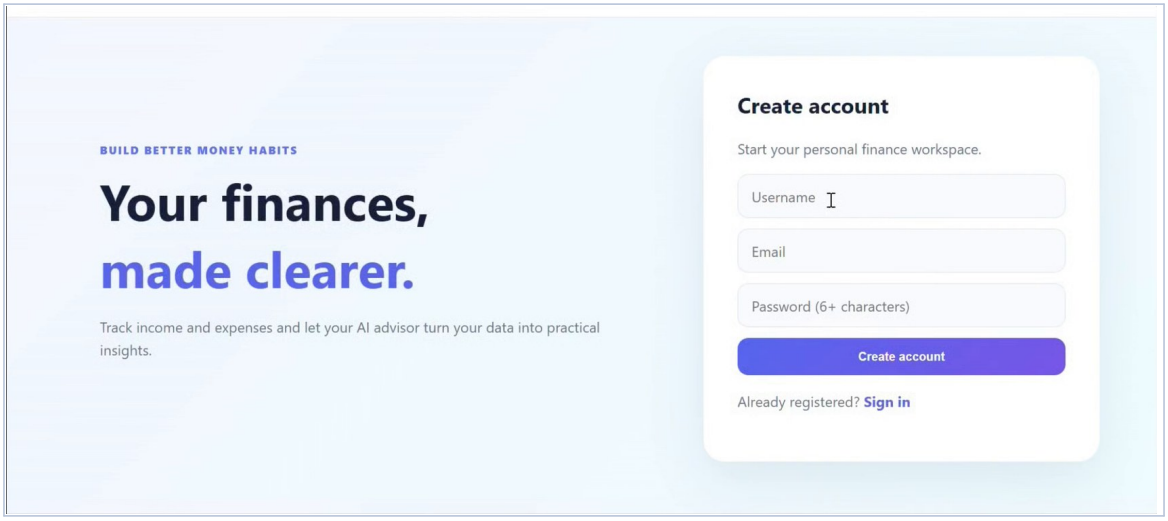


Figure 1 — Registration page

The "Create account" card asks for a username, an email and a password of at least six characters, beside the headline "Your finances, made clearer."

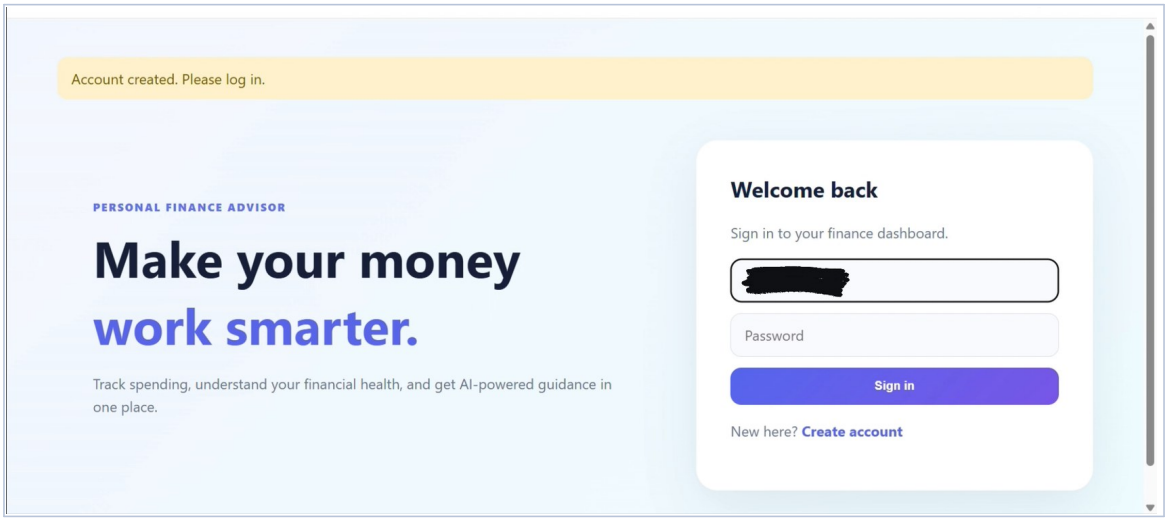


Figure 2 — Sign-in page after registration

After creating an account, the user sees the banner "Account created. Please log in." above the "Welcome back" sign-in card.

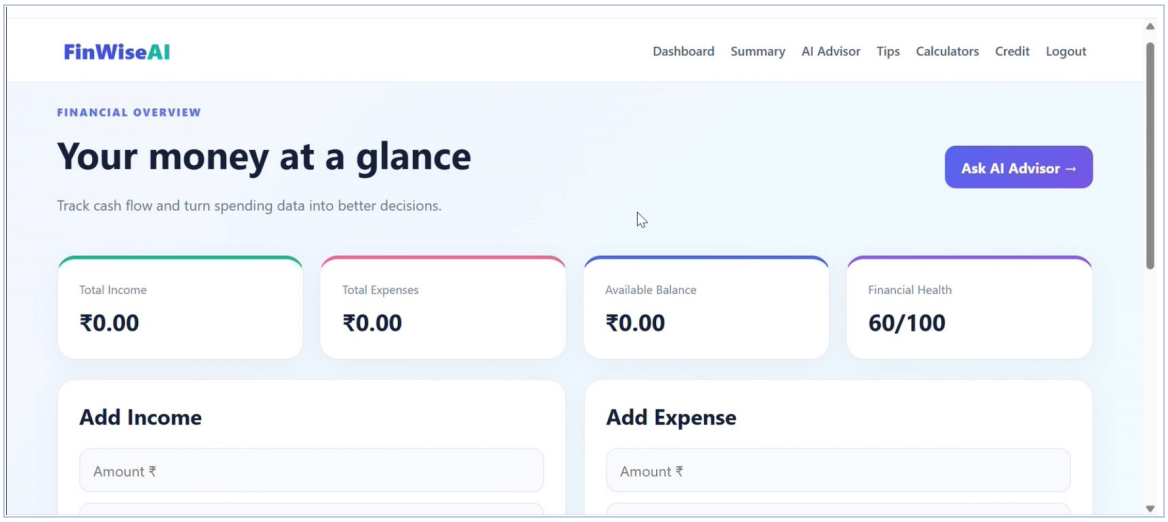


Figure 3 — Dashboard for a new account

The "Your money at a glance" dashboard before any entries: Total Income, Total Expenses and Available Balance are ₹0.00 and Financial Health is 60/100. The Ask AI Advisor button sits at the top right.

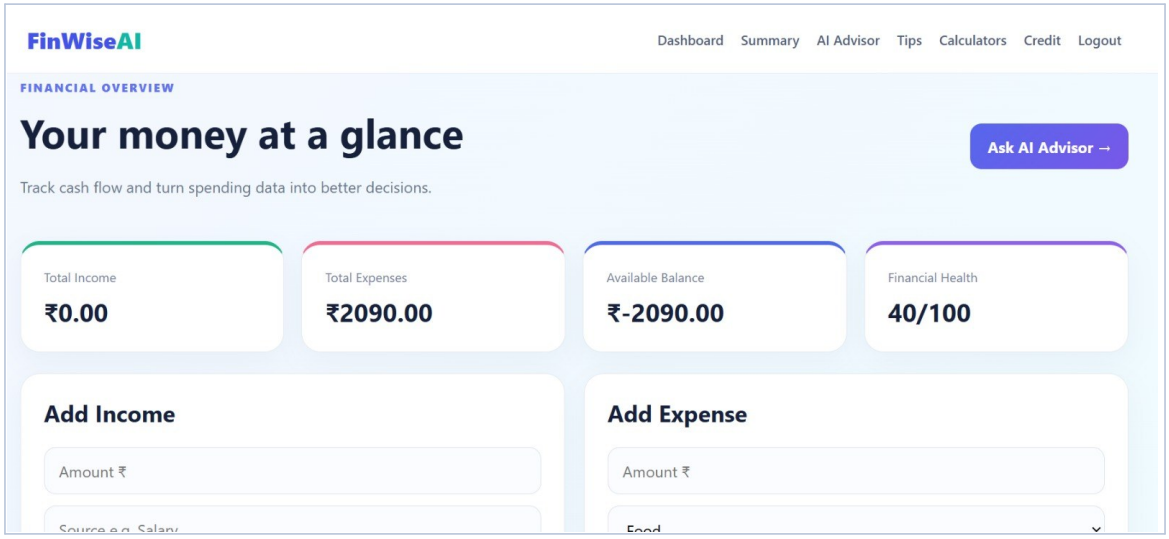


Figure 4 — Dashboard with recorded expenses

With ₹2,090 of expenses and no income, the cards show Total Expenses ₹2090.00, Available Balance ₹-2090.00 and Financial Health 40/100, above the Add Income and Add Expense forms.

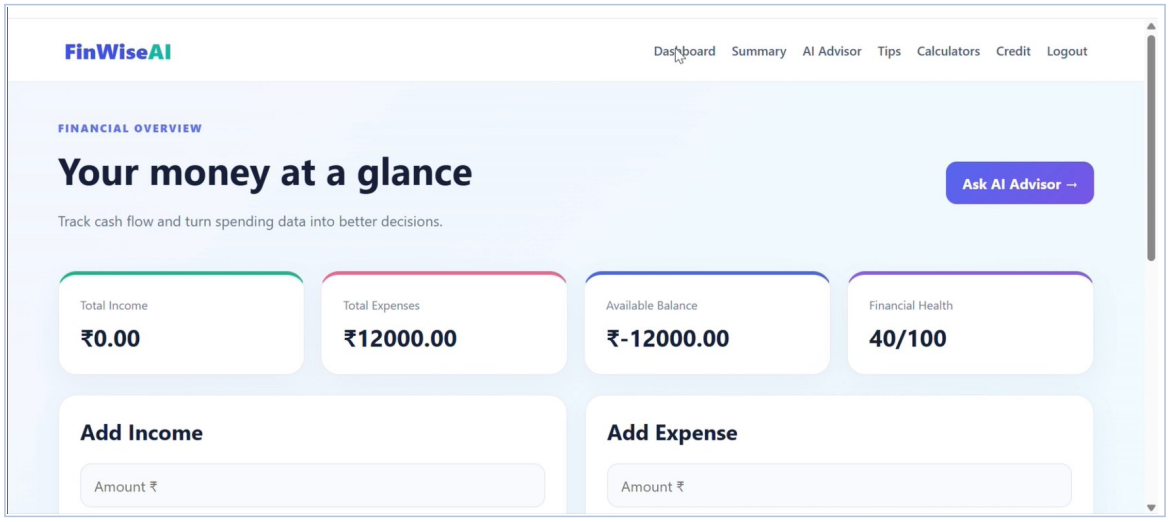


Figure 5 — Dashboard with Food, Transport and Bills expenses

After adding ₹5,000 Food, ₹4,000 Transport and ₹3,000 Bills, Total Expenses reads ₹12000.00 and Available Balance ₹-12000.00, with Financial Health at 40/100.

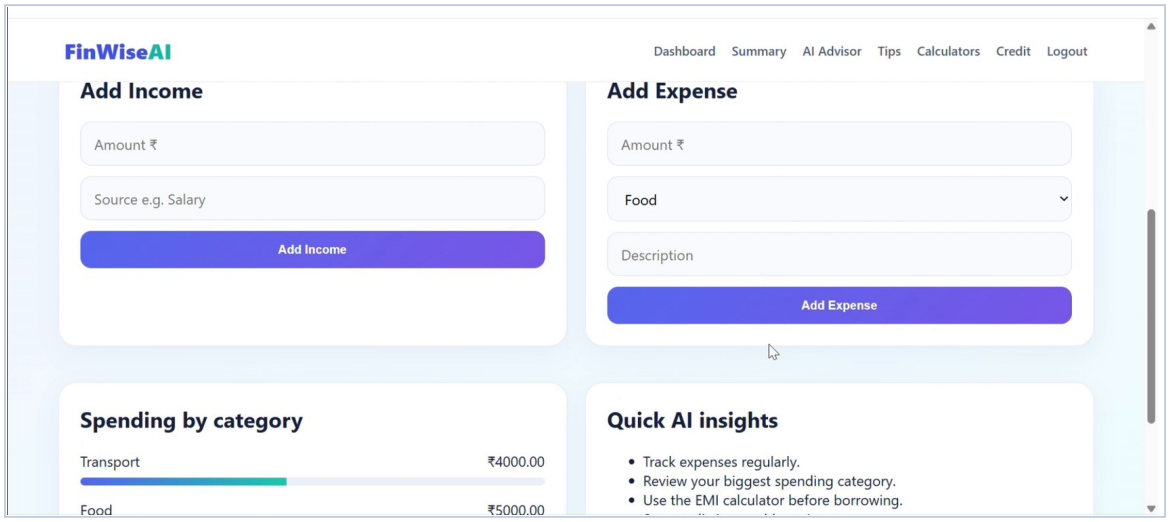


Figure 6 — Spending by category and Quick AI insights

The lower part of the dashboard lists each category with a coloured bar and amount (Transport ₹4000.00, Food ₹5000.00) next to a Quick AI insights box with short tips such as tracking expenses regularly.

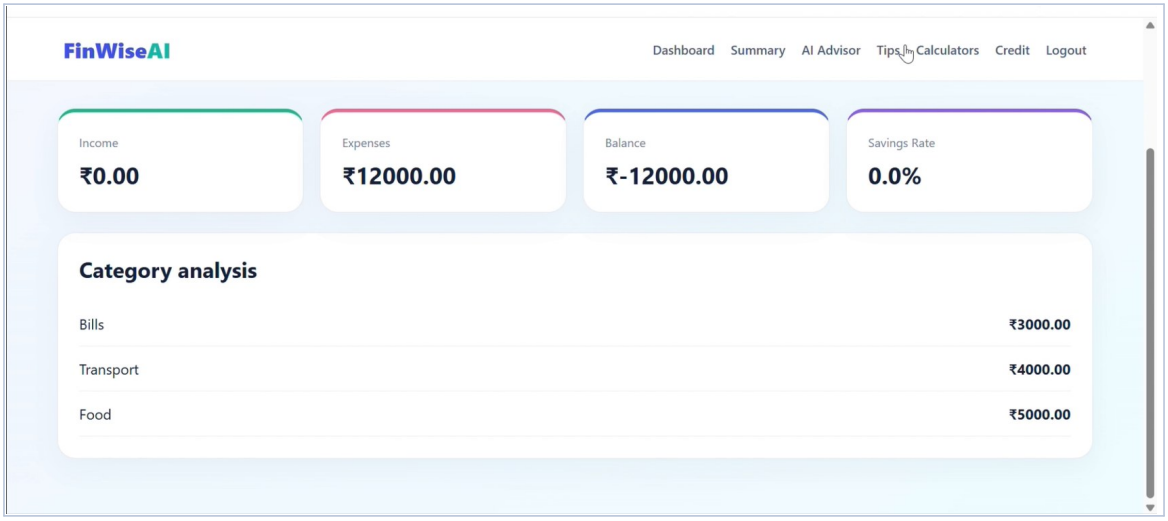


Figure 7 — Financial Summary page

The summary shows Income ₹0.00, Expenses ₹12000.00, Balance ₹-12000.00 and Savings Rate 0.0%, followed by a category analysis: Bills ₹3000.00, Transport ₹4000.00 and Food ₹5000.00.

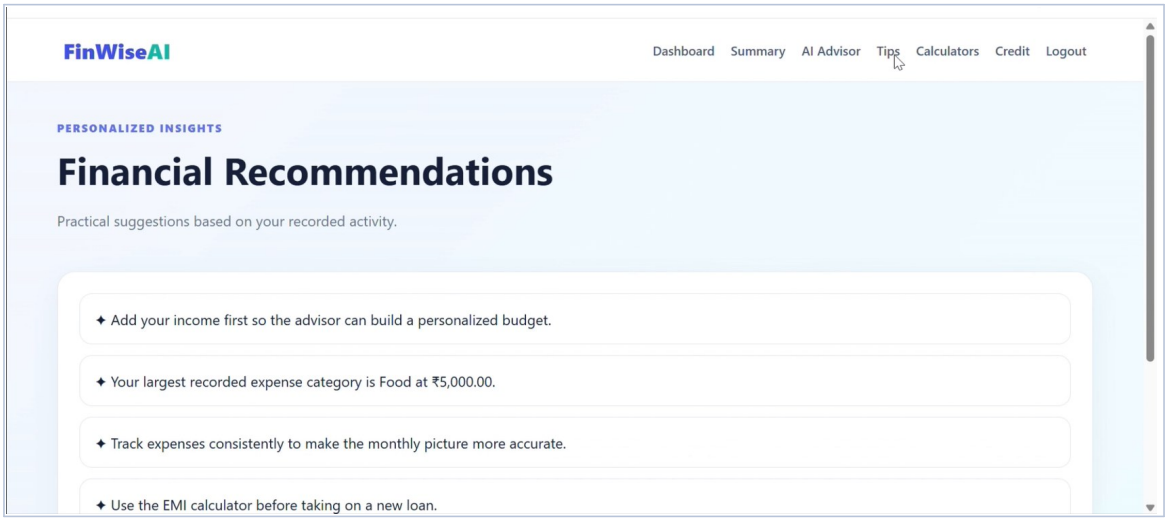


Figure 8 — Financial Recommendations page

The Tips page, titled "Personalized Insights," lists practical suggestions based on the recorded activity, such as adding income so the advisor can build a personalised budget.

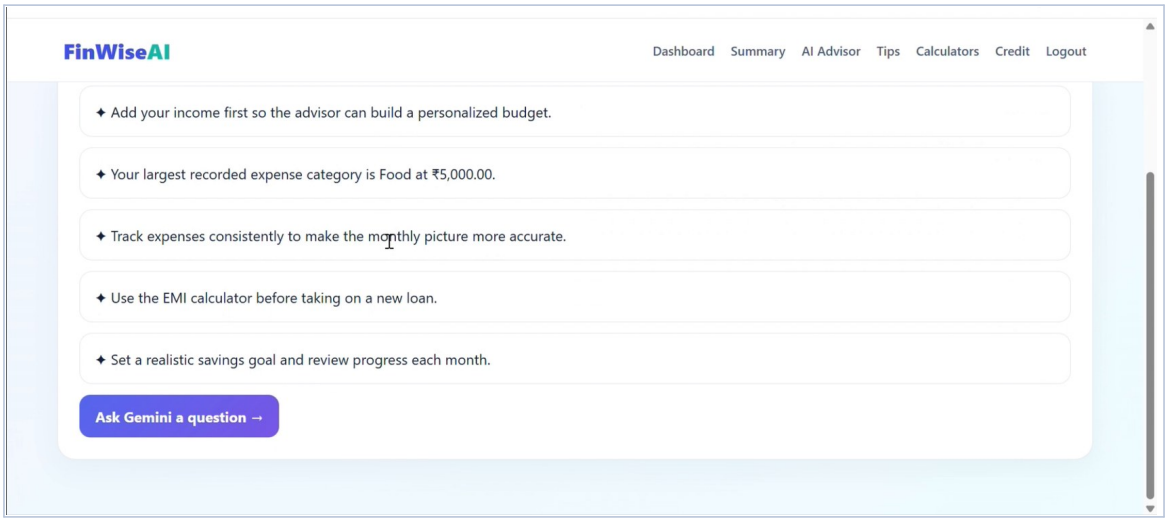


Figure 9 — Recommendations list and Ask Gemini shortcut

The rest of the list includes the largest recorded expense category (Food, ₹5,000.00), a reminder to use the EMI calculator before borrowing and a savings-goal tip, with an "Ask Gemini a question" button below.

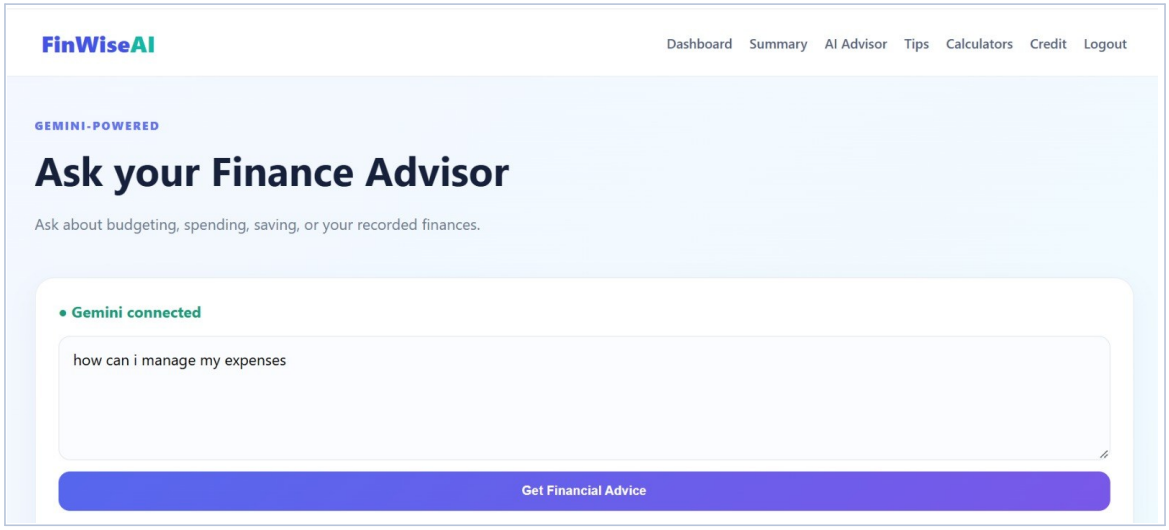


Figure 10 — AI Advisor — asking a question

The "Ask your Finance Advisor" page shows a green "Gemini connected" status, a text box holding the question "how can i manage my expenses" and the Get Financial Advice button.

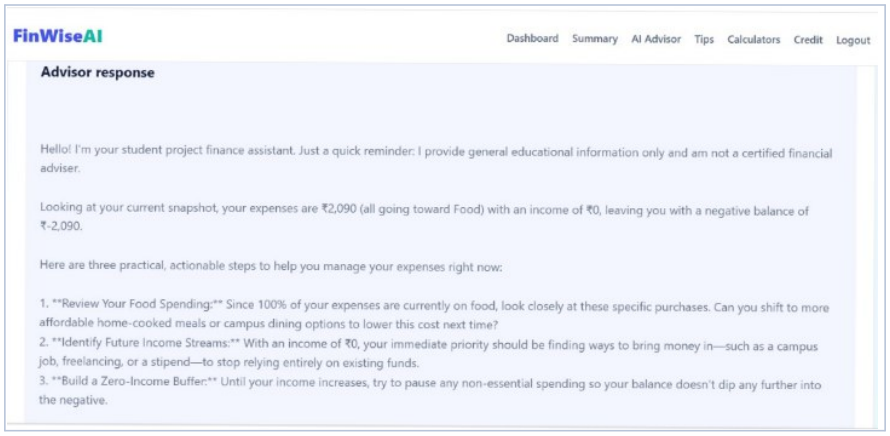


Figure 11 — AI Advisor — response using the recorded data

The reply opens with an educational disclaimer, then states expenses of ₹2,090 (all Food), income of ₹0 and a balance of ₹-2,090, followed by three practical steps.

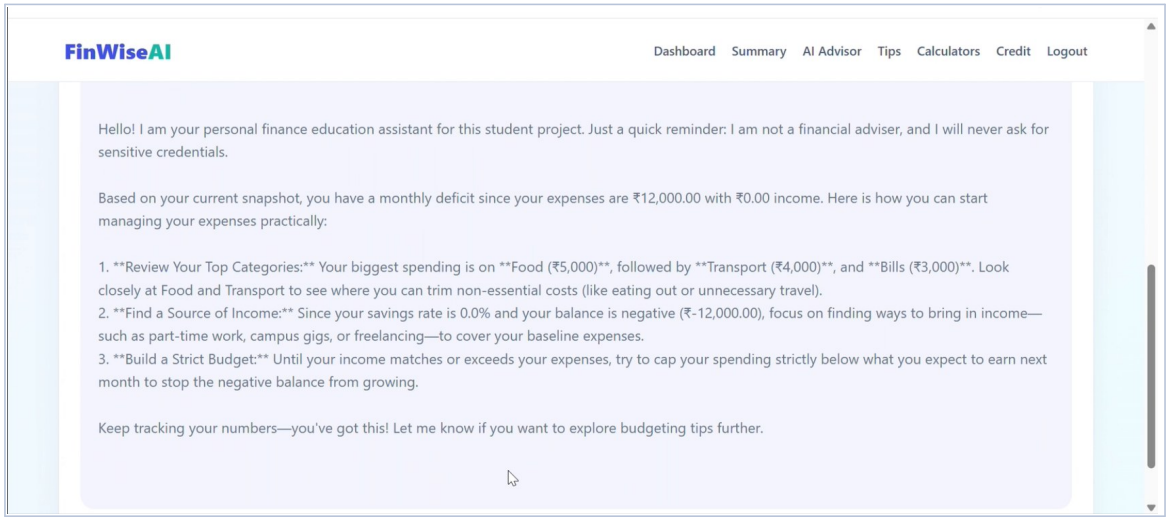


Figure 12 — AI Advisor — response for the ₹12,000 expense profile

For the Food, Transport and Bills profile the advisor names the biggest spending categories, notes the 0% savings rate and suggests finding income, reviewing top expenses and building a strict budget.

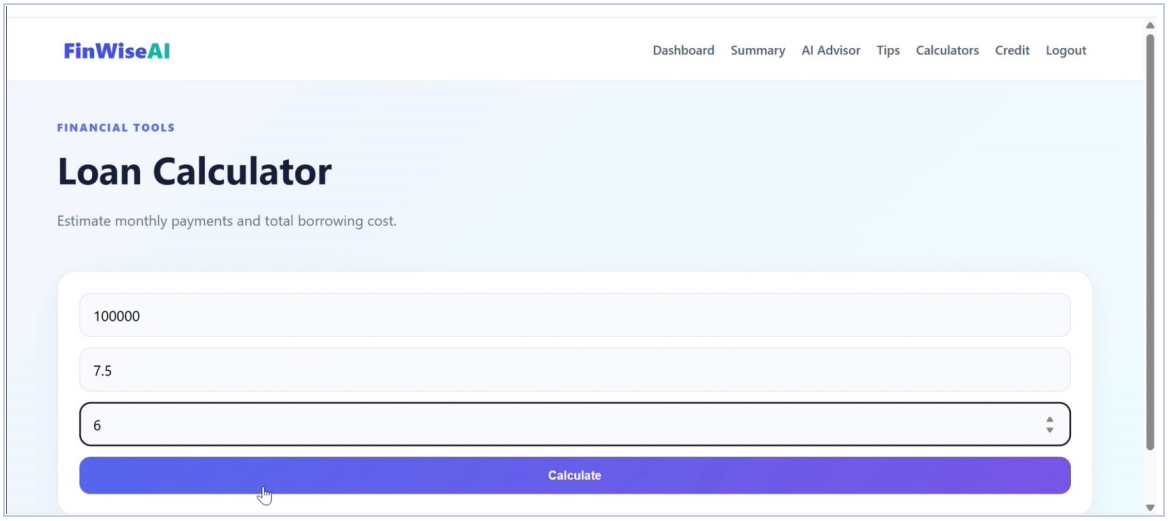


Figure 13 — Loan Calculator — input

The Loan Calculator (Financial Tools) takes the loan amount, the annual interest rate in percent and the number of years; here ₹1,00,000, 7.5% and 6 years are entered.

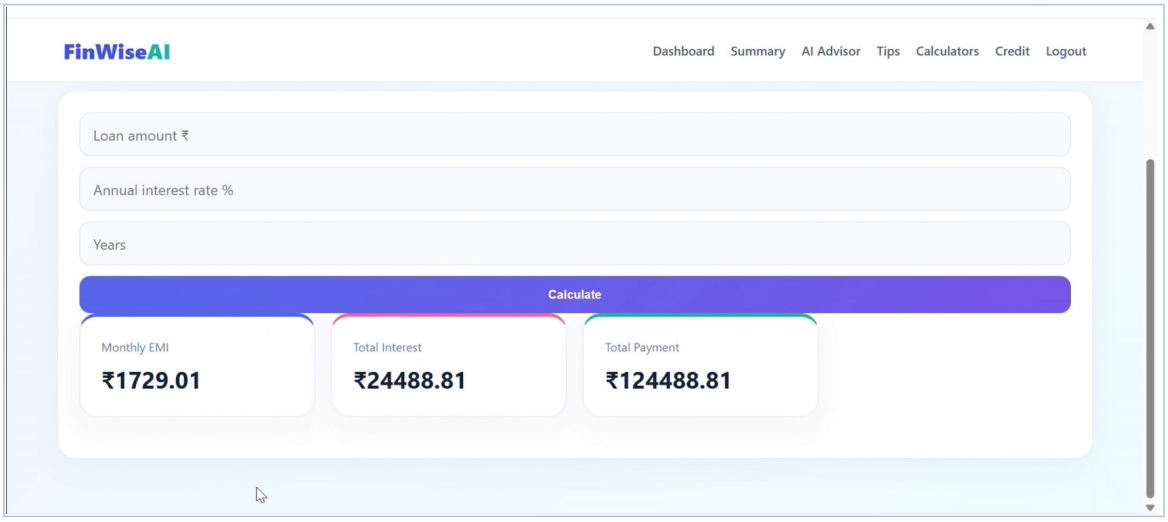


Figure 14 — Loan Calculator — result

After pressing Calculate, three cards show Monthly EMI ₹1729.01, Total Interest ₹24488.81 and Total Payment ₹124488.81, which matches the standard EMI formula.

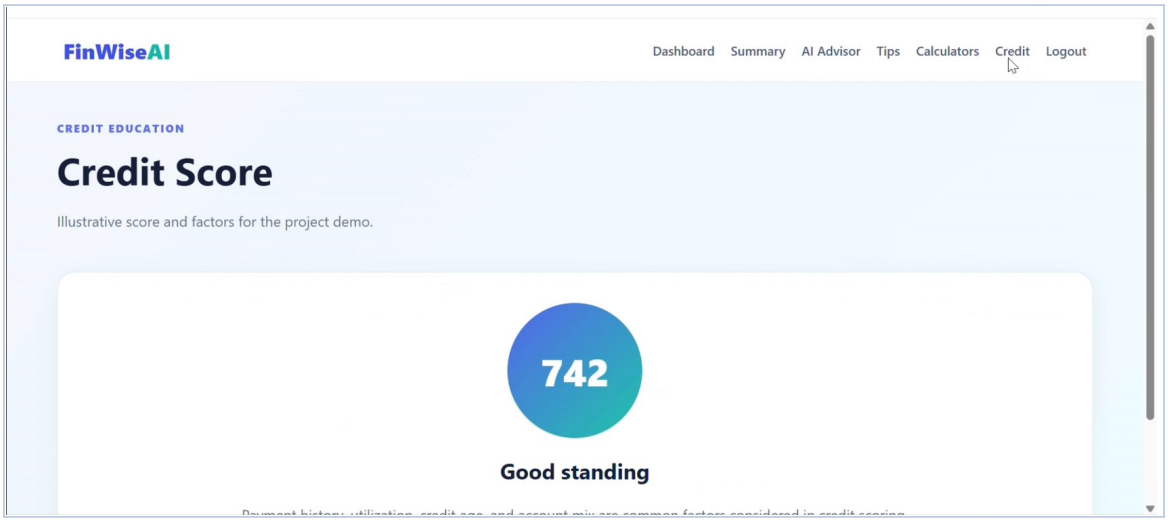


Figure 15 — Credit Score page

The Credit Education page shows an illustrative score of 742 inside a gradient ring with the label “Good standing,” and explains that payment history, utilisation, credit age and account mix are common scoring factors.

15. Security, Privacy and Limitations

Security and privacy features

- Account-based access: users register and sign in, the navigation bar offers Logout, and the password field hides what is typed and asks for at least six characters.
- The AI Advisor states in every reply that it gives general educational information and will not ask for sensitive credentials.

Recommended practices for deployment

- Store passwords only as secure hashes and keep the Gemini API key in an environment variable, never in the source code.
- Serve the application over HTTPS when it is deployed publicly, not only on a local or tunnelled address.
- Clean or safely render AI-generated text before showing it in a page.

Known limitations

- The score on the Credit page is illustrative and is not fetched from a credit bureau.
- AI Advisor replies are generated by a language model, can vary between runs and may contain mistakes; they are educational only and are not financial advice.
- In the AI Advisor replies, emphasised text appears with raw double asterisks (for example around step titles) because the reply is shown as plain text.
- Amounts are shown in a plain format such as ₹12000.00 and ₹-12000.00, without Indian digit grouping such as ₹12,000.00.
- The AI Advisor needs an internet connection and a valid API key to work.
- Testing so far is one recorded session; input validation, mobile layout and error cases still need formal testing.

16. Future Scope

- Show AI replies with proper formatting (bold text and numbered lists) instead of raw symbols.
- Add charts, such as a monthly income-versus-expense graph and a category pie chart.
- Add budgets and savings goals with progress tracking and alerts when a category goes over its limit.
- Add a transaction history page with search, filters, edit and delete, and a monthly report that can be downloaded as a PDF.
- Connect the Loan Calculator to the financial health score by comparing the new EMI with income.
- Calculate the Credit page score from user inputs instead of showing an illustrative value.
- Use Indian number formatting for all amounts and add multi-currency support.
- Add a forgot-password flow, input limits with clear error messages and automated tests.
- Deploy on a cloud platform with HTTPS and a production-grade database.
- Add voice input and regional-language support for the AI Advisor.

17. Conclusion

FinWiseAI combines expense tracking, a financial health score, summaries, recommendations, an AI advisor, a loan calculator and a credit education page in one consistent, easy-to-read application. In the recorded demonstration the modules worked together, the numbers agreed across the dashboard, summary and recommendations, the Gemini advisor gave answers based on the user's own data, and the Loan Calculator's result matched the standard EMI formula. With the improvements listed under Future Scope, such as charts, budgets, formatted AI replies and a data-driven credit page, the project can grow into a more complete financial awareness tool for students.

18. References

- Flask documentation: https://flask.palletsprojects.com
- SQLAlchemy documentation: https://docs.sqlalchemy.org
- Jinja documentation: https://jinja.palletsprojects.com
- Google Gemini API documentation: https://ai.google.dev/gemini-api/docs
- Ngrok documentation: https://ngrok.com/docs
- MDN Web Docs, HTML, CSS and JavaScript: https://developer.mozilla.org